Learning to Reach Goals via Diffusion

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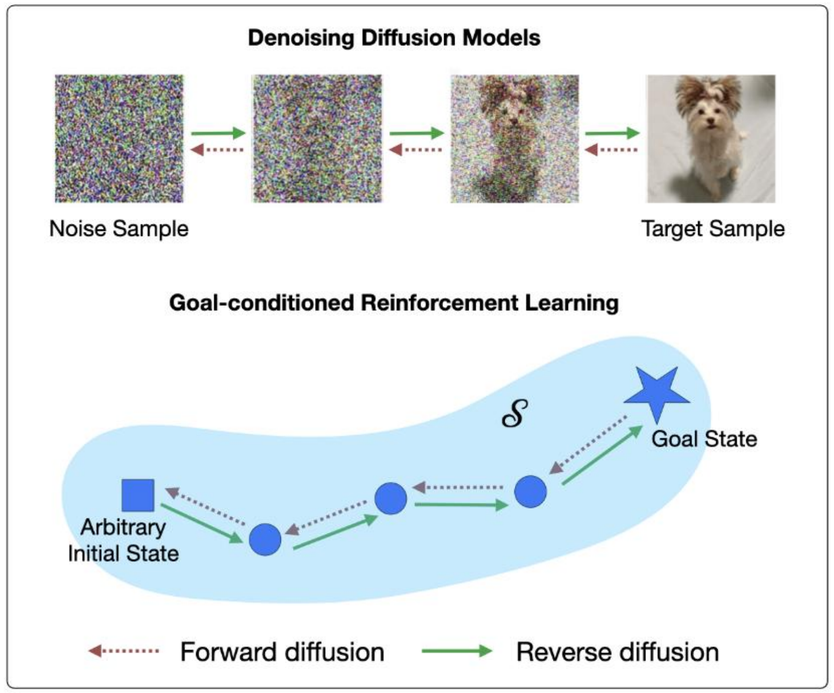
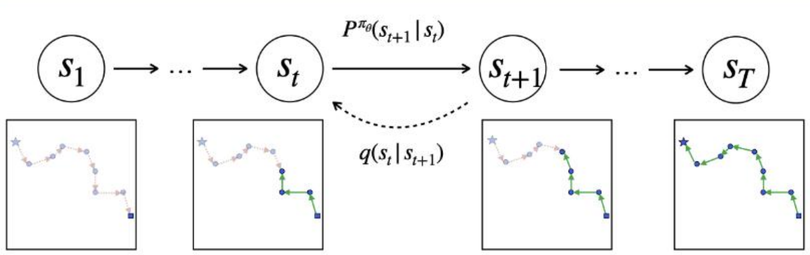
Generate trajectories that "diffuse away" from potential goals, and train a policy to reverse them



Goal Conditioned RL as Diffusion

Diffusion models: map any point from Gaussian → data manifold
Goal-conditioned RL: learn optimal path from any state → goal state

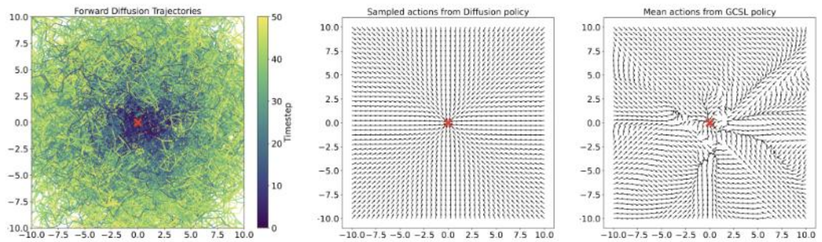


diffusion model	GCRL (Merlin)
target dataset	goal states
score function	policy
forward process	$q(s_t\|s_{t+1})$
reverse process	$\mathcal{P}^\pi(s_{t+1}\|s_t)$

Actions that lead us away from the goal in place of Gaussian noise; Policy that reverses these actions in place of score function

A Simple 2D Navigation Problem

- **Task:** Navigate to goal X; **States:** (x, y); **Actions:** unit length displacement in (x, y)
- **Forward process:** Take random actions starting from goal
- **Reverse process:** Train policy to reverse the forward trajectories

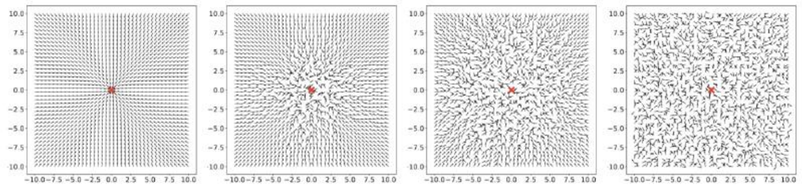


(a) Visualization of trajectories starting from the goal X generated during the forward process, (b) Predicted actions from Merlin policy, (c) Predicted actions from GCSL policy

Backward view of diffusion gives us control over the goal distribution

Effect of Time Conditioning

- Analogous to the score function in diffusion, the Merlin policy is conditioned on the time
- This time indicates how "noisy" current state is, or how far we are from the target goal state
- Policy takes optimal action far away from goal and becomes uncertain as it approaches the goal; this uncertainty appears farther from goal as time horizon increases



(a) h=1 (b) h=5 (c) h=10 (d) h=20

The time conditioning affects policy uncertainty based on expected number of steps to reach the goal

Optimization Objective

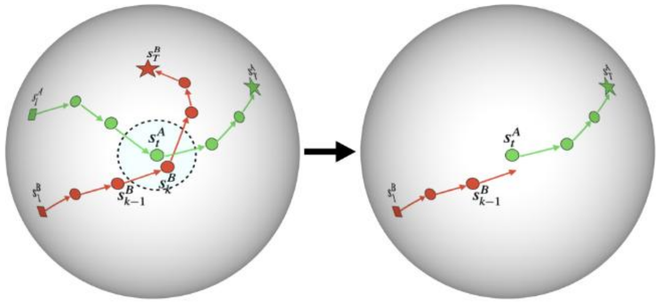
- Given an offline dataset $\mathcal{D}$, we can view the trajectories as forward diffusion chains generated by some unknown policy
- As per below Theorem, behavior cloning maximizes the probability of reaching the goal states

$$\theta^* = \arg\max_\theta \mathbb{E}_{g\sim p(g),(s,a)\sim D(g)}[\log \pi_\theta(a \mid s, g)]$$

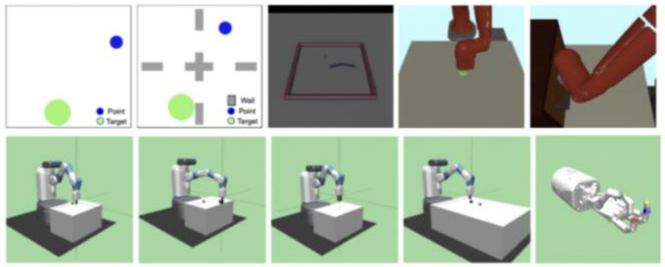
Theorem: Given a dataset $\mathcal{D}$ and target goal distribution $p(g)$, behavior cloning using a goal-conditioned policy maximizes a lower bound on the log-likelihood of the goal states $L = \mathbb{E}_{g\sim p(g), s_T\sim q(s_T|g)}[\log p_\theta(s_T)]$.

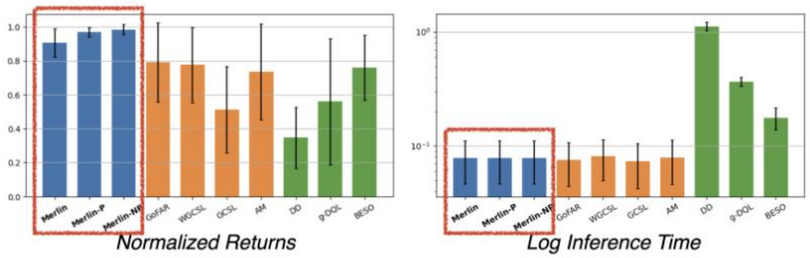
Constructing the Forward Process

- **Merlin:** Use pre-collected offline trajectories, train policy using behavior cloning
- **Merlin-P:** Learn reverse dynamics model to simulate trajectories backward from goal
- **Merlin-NP:** Use proposed trajectory stitching method to construct backward trajectories



Experimental Results





Merlin outperforms previous methods, while being highly-efficient!

Conclusions and future work

- Merlin constructs trajectories that "diffuse away" from goals and trains a policy to reverse them
- Simple training dynamics indicate potential for scalability, as proven by denoising diffusion models
- Directions for future work include extending to online setting and partially observable environments

paper code